

Incidence, Trends, and Racial/Ethnic Disparities in Female Breast Cancer in the United States, 2000-2023: A SEER Based Study

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Abstract

Introduction: In 2026, an estimated 321,910 new cases of invasive breast cancer will be diagnosed. However, there is a lack of comprehensive data on breast cancer incidence trends and how they vary across different age groups and racial and ethnic populations. Existing studies have evaluated breast cancer incidence trends over short periods, resulting in a lack of clear information on specific population variations over extended periods to ascertain disease-specific trends. **Methods:** We used the 2000-2023 Surveillance, Epidemiology, and End Results program dataset to analyze routine data and published cancer cases to identify incidence trends and disease burden by age and race/ethnicity using Joinpoint regression. We included new cases of breast cancer identified using International Classification of Diseases (ICD) codes. Patient's demographic information was obtained from the medical records. **Results:** The overall breast cancer incidence among women was 190.8 per 100,000, showing an increasing trend from 2000 to 2023. There was also a significant increase in age-adjusted incidence rates in breast cancer trends from 2000 to 2023 (AAPC = 0.76%), with the highest incidence being reported in women aged 75–79 years (456.3 per 100,000). Non-Hispanic white women recorded the highest incidence over the period (228.9 per 100,000). Non-Hispanic White women experienced increasing incidence trends between 2000 and 2023 (AAPC = 0.76%). **Conclusion:** From 2000 to 2023, there was a significant increase in breast cancer incidence trends in all groups, with notable rates among women in their 40s and above. These trends underscore the need for adaptive screening strategies and policy recommendations to initiate mammography at age 40. Progress could also be accelerated by understanding the associations among ethnic, racial, and other existing social disparities to ensure early diagnosis and treatment.

Keywords

breast cancer, burden, epidemiology, incidence; SEER; United States

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Introduction

Breast cancer is a major public health problem, being the second most common cancer globally, and accounting for 2.3 million cases and 670,000 mortalities among women in 2022.¹ In the United States (US), it is the most prevalent diagnosis and the primary cause of cancer-related deaths.² Women in the US are at a 13% risk of developing breast cancer, out of which a

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2.3% mortality is expected from the positive diagnosis.³ As a complex and chronic disease, breast cancer poses a significant burden on individuals, families, healthcare systems, and society. Although there has been progress through early diagnosis and treatment, the overall incidence remains high, underscoring the need for a multifaceted approach to its proper management.⁴

According to the American Cancer Society (ACS), there has been a rising trend in the incidence of breast cancer by 1% yearly, with 62 years being the median age of diagnosis, and a sharp incidence (16%) reported among women below 50 years.² Previous studies recorded higher incidence rates among White women (133.7 per 100,000), with high mortality rates (38% higher) reported among Black women.³ Age-adjusted rates have also been on the rise in new cancer incidences, with an average 0.6% increase annually.²

The recent shift in breast cancer patterns underscores the importance of conducting longitudinal assessments of incidence trends that span a wide range of population subgroups.⁵ The national cancer trends analysis over time provides critical information about epidemiological changes in breast cancer and the success of public health responses.⁶ Although mortality rates have decreased, changes in incidence suggest dynamic risk factors that warrant further examination.⁷ Early detection programs should be established in a manner that is relevant to the local environment and considers the country's health system capabilities.⁷ Among older populations, the public health benefits are unevenly distributed across racial, ethnic, and age-specific groups. Incidence rates, especially in younger women, and trends in Black, Hispanic, and Asian women differ significantly when compared to non-Hispanic White women.⁸ Socioeconomic inequalities, disparities in access to care, and unequal adoption of screening technology also exacerbate the differences.⁹

Between 2010 and 2019, there was an annual increase in breast cancer incidence of 0.5% yearly, with the highest increase rate (2.1%) reported among non-Hispanic Asian females.¹⁰ A similar study on incidence trends from 2001 to 2019 among women aged 65 years and older showed a significant variation in age and ethnicity, with annual percentages being higher in Hispanic women.¹¹ The majority of recent studies have evaluated breast cancer occurrence over short periods and have employed aggregate models that conceal significant variations within subgroups.^{4,10-13} There is limited data linking 2000 to 2023, a two-decade period that can provide detailed information on the changing trends in breast cancer. Other analyses have focused on specific age groups, i.e., those 65 years and above, overlooking trends in the other groups.¹⁴ The studies also lack adequate stratification against demographic variables, particularly in addressing structural health inequities.

During the COVID-19 pandemic, the suspension of routine screening services introduced additional variability into cancer surveillance data.¹⁵ Recent patterns of incidence have probably been influenced by changes in healthcare-seeking behavior, delays in diagnosis, and the discontinuation of routine screening programs after 2020.¹⁵ To determine whether changes after 2000 have affected breast cancer in any way not previously recorded, a thorough investigation of those trends is necessary. A national, population-based study of breast cancer incidence over a two-decade frame may provide essential insights into the improvements and failures of existing public health strategies. By disaggregating data on incidence by age and race/ethnicity, a more accurate picture of the disease burden and its trend direction can be obtained. We used Joinpoint regression modelling to project the levels of breast cancer incidence in US women between 2000 and 2023. Its analytical approach will highlight trends and measure changes in percentages annually, allowing the identification of specific populations with significant incidence rates and improving epidemiological understanding.

Awareness of long-term trends in the incidence of breast cancer is a primary point of reference in the development of specific screening guidelines and resource investment in prevention and early detection.¹⁶ Comprehensive trend analysis also facilitates a fairer approach to cancer control by identifying where the disparities persist or are increasing.¹⁶ Public health campaigns, policy initiatives, and clinical guidelines should be based on factual and up-to-date data that considers both past trends and the emerging changes.¹⁶ A long-term focus and hierarchical analysis will position this study favorably to satisfy these needs, while also providing quality, evidence-informed decisions at the local, state, and national levels. Unlike prior surveillance reports and analyses covering shorter study periods or limited subgroups, the present study provides a comprehensive 24-year population-based analysis of breast cancer incidence trends simultaneously stratified by age group, race/ethnicity, and sex. By applying Joinpoint regression to identify statistically significant trends within each demographic subgroup and conducting sensitivity analyses to formally assess the impact of the COVID-19 pandemic disruptions, our study offers methodological and epidemiological insights beyond what is routinely reported by professional organizations or prior analyses.

Methods

Data Source

We analyzed the Surveillance, Epidemiology, and End Results (SEER) Program dataset of the National Cancer Institute (NCI).¹⁷ For this study, we used de-identified publicly available data, and therefore, a formal ethics approval was not

required. The SEER program is a federally funded, population-based, authoritative source on cancer incidence and survival covering 48% of the US population.¹⁷ The data set routinely collects and publishes cancer cases from cancer registries, including patient demographic information, primary tumor site, tumor morphology, stage at diagnosis, first course of treatment, and follow-up for vital status.¹⁷ It encompasses cancer incidence data collected from central cancer registries in four states, as reported to the SEER program.¹⁷ It includes information on all new cancer diagnoses sourced from patient records at various medical facilities, including hospitals, physicians' offices, therapeutic radiation centers, freestanding surgical centers, and pathology laboratories.¹⁷ Central cancer registries compile these data and utilize state vital records to gather information on any cancer-related deaths that may not have been reported as cases. To ensure consistency and accuracy, the registries employ uniform data items and codes as established by the North American Association of Central Cancer Registries (NAACCR). This comprehensive dataset is submitted annually to the Centers for Disease Control and Prevention (CDC) and the NCI, meeting the US Cancer Statistics publication criteria, which attests to the high quality of the data. The population-based study was conducted in the United States using SEER registry data and included all newly diagnosed female breast cancer cases recorded from 2000 to 2023, with data collection occurring through routine cancer surveillance during the study period and no participant recruitment or prospective follow-up. The reporting of this study conforms to the Reporting of studies Conducted using Observational Routinely-collected health Data (RECORD) statement.¹⁸

Case Definitions and Tumor Characteristics

We included new cases of breast cancer identified using the International Classification of Diseases for Oncology (ICD-O)-Third Edition and histology codes. To further classify breast cancer in tumor or cancer registries, we selected them using the topography (site) and the histology (morphology) codes.

The relevant ICD-O-3 topography codes (sites) are in the C50 series, specifically: C50.0 (Nipple/Areola), C50.1 (Central portion), C50.2 (Upper-inner quadrant), C50.3 (Lower-inner quadrant), C50.4 (Upper-outer quadrant), C50.5 (Lower-outer quadrant), C50.6 (Axillary tail), C50.8 (Overlapping lesion), and C50.9 (Breast, NOS).

The histology (morphology) codes are in the "C50" range and include 8500 (ductal/NOS), 8520 (lobular), 8522 (ductal/lobular), 8480 (mucinous), 8211 (tubular), 8501 (comedocarcinoma), 8530 (inflammatory), 8510 (medullary), and 8050 and 8503 (papillary) [Details on breast cancer codes: <https://breast-cancer-research.biomedcentral.com/articles/10.1186/bcr1352/tables/1>].¹⁹

Demographic Characteristics

Based on the patient's demographic information abstracted from their medical records, we identified characteristics such as the year of cancer diagnosis ranging from 2000 to 2023 and the age groups classified into 20–24, 25–29, 30–34, 35–39, 40–44, 45–49, 50–59, 60–64, 65–69, 70–74, 75–79, 80–84, 85–89 and 90+ years. Race/Ethnicity was categorized into Hispanic, non-Hispanic Black, non-Hispanic White, American Indian/Alaska Native, and Asian/Pacific Islander.

Definitions of Cancer Incidence and Burden

The US NCI SEER program defines the cancer incidence rate as the number of new cancers of a specific site or type that occur in a given population during a given year, expressed as new cancer cases per 100,000 people at risk.

$$\text{Incidence rate} = (\text{New Cancers} / \text{Population}) \times 100,000$$

The numerator of the incidence rate is the number of new cancers, and the denominator is the population size. The number of new cancers may include multiple primary cancers in a single patient. The primary site reported is the site of origin and not the metastatic site. In general, the incidence rate would not include recurrences. The population used depends on the rate to be calculated. For cancer sites that occur in only one sex, the sex-specific population (e.g., females for breast cancer) is used.

Statistical Analysis

We calculated the incidence rates using the SEER Stat version 8.3.5. Person-years were calculated by aggregating population sizes over calendar years. We age-adjusted the incidence to the 2000 US standard population and reported it as per 100,000 person-years. Incidence trends were analyzed using the National Cancer Institute's Joinpoint Regression Analysis tool (version 4.7.0). Average annual percent changes (AAPCs) and corresponding 95% confidence intervals were estimated

to summarize the trends over the study period. The AAPC is a weighted average of the APCs from the joinpoint model, with weights equal to the length of each interval. The joinpoint program applied the best-fitting log-linear regression model to find calendar years (joinpoints) with significant APC changes, resulting in the minimum number of joinpoints required to fit the data. We used the Monte Carlo permutation test to calculate the number of significant joinpoints. To compare trends, a t-test was employed for zero joinpoints and a z-test for one or more. All hypotheses were two-sided, with statistical significance set at $p < 0.05$.

To account for pandemic-related disruptions to cancer screening, we conducted a formal sensitivity analysis. We reran the Joinpoint regression restricted to 2000–2019 to examine the pre-pandemic trend estimates. A second analysis was conducted by retaining the full study period and treating the 2020 and 2021 observations as missing values in the Joinpoint regression model.

Results

During 2000–2023, a total of 2,434,040 breast cancer cases were diagnosed. Case distribution and incidence rates by age, race or ethnicity, and year of diagnosis are presented in [Table 1](#). Age-standardized incidence rates increased gradually from 2000 to 2023, from 185.2 per 100,000 women in 2000 to 214.1 per 100,000 in 2023 ([Table 1](#)). [Table 2](#) presents breast cancer case counts and age-specific adjusted incidence rates by year of diagnosis. Across the study period, incidence rates increased within all age groups, with the highest rates observed among women aged 65–79 years, followed by those aged ≥ 80 years. [Table 3](#) presents breast cancer case counts and adjusted incidence rates by year of diagnosis and race/ethnicity. Throughout the study period, non-Hispanic White women had the highest incidence rates, followed by non-Hispanic Black women. Across all age and racial/ethnic groups, incidence rates temporarily declined during the year 2020 ([Tables 2 and 3](#)).

Joint point regression estimates showed an increasing trend in breast cancer incidence from 2000 to 2023 (AAPC = 0.76%, 95% CI = 0.53% to 1.00%), with notable differences observed in specific subgroups ([Table 4](#)). The AAPC estimates were broadly similar in the primary analysis and in the COVID-19 sensitivity analyses, both for overall breast cancer incidence and across racial, ethnic, and age subgroups ([Table 4](#)). Among racial and ethnic groups, the overall direction of the trends and their statistical significance did not change across analyses. This included non-Hispanic Black women, for whom the estimates remained comparable in the primary analysis and both sensitivity analyses. Some small differences were seen in certain age groups, especially women aged 60–64, 65–69, and 75–79 years. In these groups, statistical significance varied somewhat between analyses, even though the AAPC estimates themselves were generally similar ([Table 4](#)). Because the findings were largely consistent across analyses, the primary analysis results are presented below.

Incidence Trends by Age

Age-specific trends indicate that the incidence of breast cancer increased steadily with age, with the highest incidence among women aged 75–79 years (456.3 per 100,000) ([Table 1](#)). Annual age-specific incidence rates are presented in [Table 2](#), and the corresponding joinpoint regression trends are presented in [Table 4](#) and [Figure 1](#).

Joinpoint regression analysis showed a consistent, significant increasing trend for all age groups under 45 years from 2000–2023. The 20–24 years age group had an AAPC of 0.91% (95% CI = 0.16% to 1.64%). The other groups had APC = 1.60% (95% CI = 1.36% to 1.83%) for 25–29 years, AAPC = 0.95% (95% CI = 0.72% to 1.18%) for 30–34 years, AAPC = 0.46% (95% CI = 0.22% to 0.70%) for 35–39 years, and AAPC = 0.70% (95% CI = 0.52% to 0.88%) for 40–44 years. A significant increase in breast cancer incidence was observed among women aged 45–49 years between the years 2000 and 2023 (AAPC = 0.49%, 95% CI = 0.29% to 0.68%). There was an increase in breast cancer incidence among women aged 50–54 years (AAPC = 0.13%, 95% CI = -0.22% to 0.46%), although it was not significant. There was a significant decrease in breast cancer incidence among women aged 55–59 years between the years 2000 and 2023 (AAPC = -0.56%, 95% CI = -0.92% to -0.20%). A significant decrease was also observed among women aged 60–64 years between the years 2000 and 2023 (AAPC = -0.30%, 95% CI = -0.60% to -0.002%). Women aged 65–69 years showed an increase between the years 2000 and 2023 (AAPC = 0.26%, 95% CI = -0.01% to 0.53%). For women aged 70–74 years, there was an increase in breast cancer incidence between 2000 and 2023 (AAPC = 0.64%, 95% CI = 0.27% to 1.0%). There was a significant increase in breast cancer incidence among women aged 75–79 years between the years 2000 and 2023 (AAPC = 0.30%, 95% CI = 0.03% to 0.58%).

There was an increasing trend for 80–84 and 85–89 years for 2000–2023, with AAPC = 0.08% (95% CI = -0.26% to 0.40) and AAPC = -0.09% (95% CI = -0.35 to 0.16%) respectively, although it was not significant. The 90+ years group showed a significant decrease (AAPC = -0.86%, 95% CI = -1.25% to -0.49%) for the years 2000 to 2023 ([Table 4](#); [Figure 1](#)).

Table 1. Breast Cancer Age-Standardized Rates Among US Women by Age at Diagnosis, Race/Ethnicity, and Year of Diagnosis During 2000-2023 in the United States

Variable	Cases (%)	Incidence rate per 100,000 (95% CI)
Age Group		
20–24 years	1,894 (0.1%)	1.6 (1.5, 1.7)
25–29 years	11,476 (0.5%)	9.6 (9.4, 9.8)
30–34 years	34,384 (1.4%)	28.7 (28.4, 29.0)
35–39 years	74,857 (3.1%)	62.7 (62.3, 63.2)
40–44 years	150,400 (6.2%)	125.9 (125.3, 126.6)
45–49 years	227,243 (9.3%)	192.6 (191.8, 193.4)
50–54 years	264,973 (10.9%)	231.1 (230.2, 231.9)
55–59 years	288,940 (11.9%)	273.4 (272.4, 274.4)
60–64 years	311,104 (12.8%)	342.0 (340.8, 343.2)
65–69 years	309,636 (12.7%)	415.4 (413.9, 416.8)
70–74 years	268,032 (11.0%)	449.7 (448.0, 451.4)
75–79 years	211,477 (8.7%)	456.3 (454.3, 458.2)
80–84 years	148,363 (6.1%)	429.6 (427.4, 431.8)
85–89 years	86,075 (3.5%)	387.4 (384.8, 390.0)
90+ years	45,186 (1.9%)	310.8 (308.0, 313.7)
Race/Ethnicity		
Hispanic	308,776 (12.7%)	116.6 (116.2, 117.0)
American Indian/Alaska Native	10,647 (0.4%)	134.1 (131.6, 136.7)
Asian or Pacific Islander	164,462 (6.8%)	143.8 (143.1, 144.5)
Non-Hispanic Black	270,111 (11.1%)	174.9 (174.2, 175.5)
Non-Hispanic White	1,680,044 (69.0%)	228.9 (228.5, 229.2)
Year of Diagnosis		
2000	86,810 (3.6%)	185.2 (183.9, 186.4)
2001	87,908 (3.6%)	185.1 (183.9, 186.4)
2002	87,919 (3.6%)	183.0 (181.8, 184.2)
2003	84,745 (3.5%)	174.4 (173.3, 175.6)
2004	85,538 (3.5%)	174.3 (173.2, 175.5)
2005	86,675 (3.6%)	175.1 (174.0, 176.3)
2006	88,552 (3.6%)	177.0 (175.8, 178.1)
2007	91,726 (3.8%)	181.3 (180.1, 182.4)
2008	93,843 (3.9%)	183.2 (182.0, 184.4)
2009	96,841 (4.0%)	186.6 (185.5, 187.8)
2010	95,321 (3.9%)	181.4 (180.2, 182.5)
2011	98,680 (4.1%)	185.4 (184.3, 186.6)
2012	100,053 (4.1%)	185.8 (184.7, 187.0)
2013	102,827 (4.2%)	189.0 (187.9, 190.2)
2014	105,272 (4.3%)	191.5 (190.3, 192.6)
2015	108,306 (4.4%)	195.1 (193.9, 196.2)
2016	108,413 (4.5%)	193.5 (192.4, 194.7)
2017	111,612 (4.6%)	197.7 (196.6, 198.9)
2018	114,479 (4.7%)	201.5 (200.4, 202.7)
2019	119,121 (4.9%)	208.5 (207.3, 209.7)
2020	108,254 (4.4%)	188.6 (187.5, 189.7)
2021	122,930 (5.1%)	213.7 (212.5, 214.9)
2022	122,886 (5.0%)	212.2 (211.0, 213.3)
2023	125,329 (5.1%)	214.1 (212.9, 215.3)

Incidence Trends by Race and Ethnicity

From 2000-2023, breast cancer incidence was highest among non-Hispanic White women (228.9 per 100,000) compared to non-Hispanic Black women (174.9 per 100,000), Asian or Pacific Islander women (143.8 per 100,000), American Indian/Alaska Native women (134.1 per 100,000), and Hispanic women (116.6 per 100,000) (Table 1). Annual incidence rates by

Table 2. Breast Cancer Case Counts and Adjusted Incidence Rates by Year of Diagnosis and Age-Group

Year	20-39		40-49		50-64		65-79		≥80	
	Count	Rate	Count	Rate	Count	Rate	Count	Rate	Count	Rate
2000	4857	25.63	14938	153.41	29129	304.39	27095	446.67	10791	421.75
2001	4973	26.21	14947	150.67	30073	302.95	26864	443.46	11051	423.69
2002	4741	24.99	15054	149.40	30839	299.75	26389	436.29	10896	410.05
2003	4753	25.08	15464	151.60	29601	277.68	24395	402.19	10532	389.30
2004	4788	25.32	15638	151.93	29945	271.22	24258	399.49	10909	396.34
2005	4607	24.43	15651	151.79	30792	269.56	24359	399.83	11266	401.48
2006	4656	24.58	15935	154.80	31744	268.59	25142	409.65	11075	388.60
2007	4645	24.39	16153	158.00	33483	274.04	26149	420.27	11296	390.10
2008	4808	25.06	15987	157.43	34610	275.27	27003	423.49	11435	389.72
2009	4740	24.52	16093	159.21	35707	275.60	28351	434.54	11950	402.38
2010	4666	23.99	15454	153.43	35293	264.07	28311	424.89	11597	385.29
2011	4866	24.82	15681	156.20	36844	268.32	29722	436.06	11567	382.28
2012	4828	24.40	15484	155.40	37039	266.62	31110	434.71	11592	381.35
2013	5023	25.15	15575	158.11	37927	269.31	32771	439.64	11531	379.45
2014	5030	24.92	15498	159.28	38982	273.26	34325	442.62	11437	375.59
2015	5180	25.45	15575	161.28	39876	276.84	36010	447.47	11665	381.41
2016	5316	25.90	14961	155.60	39714	274.53	36954	443.19	11468	372.94
2017	5406	26.20	15727	163.75	40129	276.97	38792	449.69	11558	374.46
2018	5462	26.38	15820	164.95	40305	278.46	40580	455.01	12312	395.92
2019	5829	28.09	16337	170.66	41488	287.19	42887	465.13	12580	401.22
2020	5564	26.84	15140	158.30	37547	260.22	38822	407.65	11181	355.06
2021	5982	29.01	16598	174.04	41784	290.54	45579	464.26	12987	408.84
2022	5845	28.24	16659	173.54	41052	287.67	46090	455.81	13240	408.70
2023	6046	28.95	17274	177.49	41114	290.34	47187	454.13	13708	407.37

race/ethnicity are presented in [Table 3](#), and the corresponding joinpoint regression trends are presented in [Table 4](#) and [Figure 2](#).

Joinpoint regression analysis showed that all the race/ethnicity groups in the study Hispanic, non-Hispanic White, Non-Hispanic American Indian/Alaska Native, non-Hispanic Asian/Pacific Islander, and non-Hispanic Black all experienced a significant increase in breast cancer incidence for the years 2000-2023. The corresponding annual percentage changes and 95% confidence intervals were (AAPC = 1.91%, 95% CI = 1.66% to 2.15%), (AAPC = 0.76%, 95% CI = 0.51% to 1.00%), (AAPC = 3.10%, 95% CI = 2.71% to 3.50%), (AAPC = 2.20%, 95% CI = 1.93% to 2.48%), and (AAPC = 1.31%, 95% CI = 1.12% to 1.49%), respectively ([Table 4](#); [Figure 2](#)).

Discussion

Evolving Incidence Trends in Breast Cancer

Data from our study found a modest yet statistically significant increase in the overall incidence of female breast cancer in the US between 2000 and 2023, with the age-standardized rate rising from 185.2 to 214.1 per 100,000 (joinpoint AAPC = 0.76%, 95% CI: 0.53% to 1.00%). This overall trend aligns with recent American Cancer Society reports documenting an increase at a steady 1% in breast cancer incidence after the decline in the early 2000s.³ The decline in incidence observed in the early 2000s, particularly in 2003 among older women, might be largely attributed to the reduced use of post-menopausal hormone therapy following the Women's Health Initiative (WHI) report.²⁰ However, after the temporary decline, breast cancer incidence increased, contributing to the net increase seen in 2023, showing that the burden of breast cancer has continued to grow in absolute terms. Even with a 0.5–1% annual increase in recent years, the cumulative impact is significant, with over 1.4 million women diagnosed over the past two decades.

The findings from our analysis are comparable with the recent global studies. For example, a recent Global Burden of Disease (GBD) 2022 analysis reported a global age-standardized breast cancer incidence rate of approximately 46 per 100,000 women in 2021. In the US, the incidence rate was 94 per 100,000 which is similar to our findings (99.2 per 100,000 women).²¹ This disparity is consistent with the higher breast cancer incidence observed in high-income countries,

Table 3. Breast Cancer Case Counts and Adjusted Incidence Rates by Year of Diagnosis and Race/Ethnicity

Year	Hispanic		Indian/Alaska native		Asian/Pacific islander		Black		White	
	Count	Rate	Count	Rate	Count	Rate	Count	Rate	Count	Rate
2000	7544	97.95	265	92.08	3530	111.05	8219	149.62	67252	222.56
2001	7599	94.92	282	96.60	3799	114.12	8265	148.40	67963	224.38
2002	8051	96.87	255	86.34	4067	117.14	8584	152.16	66962	220.83
2003	8206	95.31	302	101.03	4026	111.63	8700	152.45	63511	209.18
2004	8518	95.80	331	109.37	4355	116.64	8875	153.81	63459	208.97
2005	9152	99.79	313	102.39	4569	118.23	9037	155.68	63604	209.58
2006	9369	99.06	314	101.79	4759	119.04	9366	158.99	64744	213.09
2007	9974	102.22	354	113.63	5101	123.55	9918	166.03	66379	218.10
2008	10576	104.93	352	111.64	5427	127.31	10109	166.87	67379	220.83
2009	11233	107.86	384	120.31	5559	126.52	10698	173.75	68967	225.36
2010	11087	103.14	389	120.45	5989	132.40	10430	166.57	67426	219.62
2011	12083	109.43	456	138.98	6327	135.32	11067	173.63	68747	223.22
2012	12479	110.22	467	140.10	6590	136.63	11434	176.36	69083	223.74
2013	13461	116.14	451	133.43	7126	143.39	11785	178.84	70004	226.47
2014	13783	116.22	498	145.34	7293	142.21	12136	181.23	71562	231.21
2015	14688	121.14	504	145.31	7922	149.60	12436	182.94	72756	235.00
2016	15190	122.65	506	144.41	8040	147.46	12572	182.51	72105	233.05
2017	15819	125.26	550	155.44	8713	155.70	13043	187.06	73487	237.90
2018	16471	128.03	549	153.84	9321	163.23	13116	186.01	75022	243.43
2019	17790	135.94	622	172.88	9619	165.50	14161	198.60	76929	250.30
2020	16205	121.68	515	141.73	8739	148.15	12816	178.08	69979	228.57
2021	19091	141.16	659	180.09	11043	184.97	14238	196.67	77899	255.97
2022	19648	142.34	635	172.43	11204	182.12	14466	197.86	76933	254.02
2023	20759	146.60	694	187.01	11344	177.96	14640	197.69	77892	257.73

likely reflecting differences in reproductive and lifestyle-related risk factors, obesity prevalence, and greater access to mammographic screening and early detection programs. Furthermore, GBD estimates demonstrated increasing breast cancer incidence trends globally, with increased incidence in East and Southeast Asia, which is comparable to our findings of significant incidence increases among Asian/Pacific Islander women in the United States.

These trends highlight a growing demand for healthcare systems and emphasize the critical need for sustained, long-term strategies in cancer prevention and control.

Age-specific Trends and Generational Shifts

The difference in trends by age group is quite significant. Younger women (<45 years) experienced a consistent, significant increase in breast cancer throughout 2000–2023. By contrast, incidence among older women, particularly those 50–84 years, showed a decline in the early 2000s, followed by relative stability or modest fluctuations thereafter. For example, national cancer registry analyses have documented that incidence in women aged 50–64 years dropped markedly (a decrease of 2.8% on average per year) in 1999–2004, then increased only slightly in subsequent years to 1.5%.⁴ Our data are consistent with this trend, showing an initial decline among older cohort likely due to the reduction in hormone therapy use post-2002, which contributed to the overall decrease in incidence during the early 2000s. In contrast, younger cohorts did not experience such a decline.^{3,4} The continuous increase in the incidence among younger women may reflect generational changes in risk factor exposures and reproductive behaviors. Women born in more recent decades tend to have children at older ages or have fewer children, factors which are associated with higher breast cancer risk due to prolonged estrogen exposure and reduced lifetime breastfeeding.²² Environmental and lifestyle changes, for example, greater alcohol consumption or other potential unidentified exposures, may also contribute to the elevated breast cancer risk observed among younger women.²³ In contrast, the older generations experienced certain risk reductions (e.g., the restriction of hormone replacement therapy around 2002) that resulted in an immediate decline in incidence, especially for estrogen receptor-positive tumors.²⁴ Following this period, breast cancer incidence among older women has remained relatively stable, suggesting that factors such as screening saturation and possibly cohort effects may have limited further increase in these age groups. However, it is important to note

Table 4. Comparison of Joinpoint Regression Trend Estimates Across Primary and COVID-19 Sensitivity Analyses for Female Breast Cancer Incidence in the United States, 2000–2023

Subgroup	Primary analysis 2000–2023 (all years) AAPC (95% CI)	Sensitivity 1 2000–2019 (Pre-COVID) AAPC (95% CI)	Sensitivity 2 2000–2023 (excl. 2020–2021) AAPC (95% CI)
Overall	0.76 (0.53 to 1.00)	0.67 (0.39 to 0.95)	0.78 (0.54 to 1.03)
By Race/Ethnicity			
Hispanic	1.91 (1.66 to 2.15)	1.82 (1.57 to 2.07)	1.95 (1.71 to 2.19)
American Indian/Alaska Native	3.10 (2.71 to 3.5)	3.30 (2.79 to 3.81)	3.22 (2.83 to 3.62)
Asian or Pacific Islander	2.20 (1.93 to 2.48)	2.13 (1.86 to 2.39)	2.22 (1.99 to 2.46)
Non-Hispanic Black	1.31 (1.12 to 1.49)	1.45 (1.28 to 1.61)	1.38 (1.24 to 1.52)
Non-Hispanic White	0.76 (0.51 to 1.00)	0.67 (0.38 to 0.96)	0.78 (0.53 to 1.03)
By Age Group			
20–24 years	0.91 (0.16 to 1.64)	1.20 (0.32 to 2.08)	0.84 (0.08 to 1.64)
25–29 years	1.60 (1.36 to 1.83)	1.59 (1.26 to 1.90)	1.55 (1.30 to 1.81)
30–34 years	0.95 (0.72 to 1.18)	0.89 (0.58 to 1.20)	0.94 (0.69 to 1.20)
35–39 years	0.46 (0.22 to 0.70)	0.32 (0.08 to 0.56)	0.44 (0.21 to 0.67)
40–44 years	0.70 (0.52 to 0.88)	0.58 (0.42 to 0.74)	0.72 (0.55 to 0.89)
45–49 years	0.49 (0.29 to 0.68)	0.29 (0.11 to 0.46)	0.48 (0.28 to 0.70)
50–54 years	0.13 (-0.22 to 0.46)	-0.11 (-0.49 to 0.27)	0.13 (-0.20 to 0.46)
55–59 years	-0.56 (-0.92 to -0.20)	-0.84 (-1.36 to -0.32)	-0.60 (-1.00 to -0.17)
60–64 years	-0.30 (-0.60 to -0.002)	-0.38 (-0.74 to -0.03)	-0.26 (-0.54 to 0.02)
65–69 years	0.26 (-0.01 to 0.53)	0.41 (0.09 to 0.72)	0.36 (0.12 to 0.61)
70–74 years	0.64 (0.27 to 1.0)	0.78 (0.34 to 1.22)	0.70 (0.37 to 1.04)
75–79 years	0.30 (0.03 to 0.58)	0.24 (-0.12 to 0.60)	0.32 (0.05 to 0.60)
80–84 years	0.08 (-0.26 to 0.40)	-0.14 (-0.48 to 0.18)	0.09 (-0.22 to 0.42)
85–89 years	-0.09 (-0.35 to 0.16)	-0.15 (-0.40 to 0.10)	-0.06 (-0.28 to 0.18)
90+ years	-0.86 (-1.25 to -0.49)	-1.12 (-1.50 to -0.74)	-0.87 (-1.25 to -0.46)

AAPC = Average Annual Percentage Change, representing the weighted average of APCs across all trend segments identified by the Joinpoint model.

Bold values indicate statistical significance at $p < 0.05$.

Sensitivity 1 excludes pandemic years entirely (the analysis ends in 2019).

Sensitivity 2 treats 2020 and 2021 as missing data points in the Joinpoint model; the full 2000–2023 period is modeled with those two years excluded.

that even stable incidence rates in older women translate to a growing number of cases as the population ages. The aging of the baby boomer cohort and an overall increase in life expectancy have resulted in a growing number of women reaching age ranges associated with increased breast cancer risk.²⁵ Assuming incidence rates remain constant, the number of breast cancer diagnoses among older women is projected to increase dramatically simply due to demographic shifts. One analysis conducted by Alberg and Singh et al, in 2001 projected a 72% surge in breast cancer cases among women 65 and older between 1998 and 2025 (from ~89,500 cases in 1998 to ~154,000 in 2025) if rates stayed unchanged.²⁶ Even in the absence of additional increases in age-specific incidence, this demographic driver (an aging female population) predicts a significant increase in the burden of breast cancer in the years to come. The interplay of cohort effects, lifestyle changes, and population aging has resulted in more young women developing breast cancer and a persistent high burden in older women, which together contribute to the overall upward trend in incidence and case counts.

Racial and Ethnic Disparities

Consistent with prior reports, our study identified differences in the new cases of breast cancer by race and ethnicity. The highest overall incidence was seen among non-Hispanic White women, followed by Hispanic, Asian/Pacific Islander (A/PI), American Indian/Alaska Native (AI/AN), and non-Hispanic Black women. However, the temporal trends over the 24-year period varied significantly between groups. All non-White populations showed a greater relative increase in incidence than White women, effectively narrowing the historic gap. The highest rise in incidence was observed among A/PI women, especially those under 50, whose incidence rates now approach or exceed those of young White and Black women.^{27,28}

These changes most likely reflect increased screening availability in previously underprivileged populations, evolving reproductive patterns, and lifestyle risk factors.¹³ Nevertheless, the implications are complex. While rising incidence may

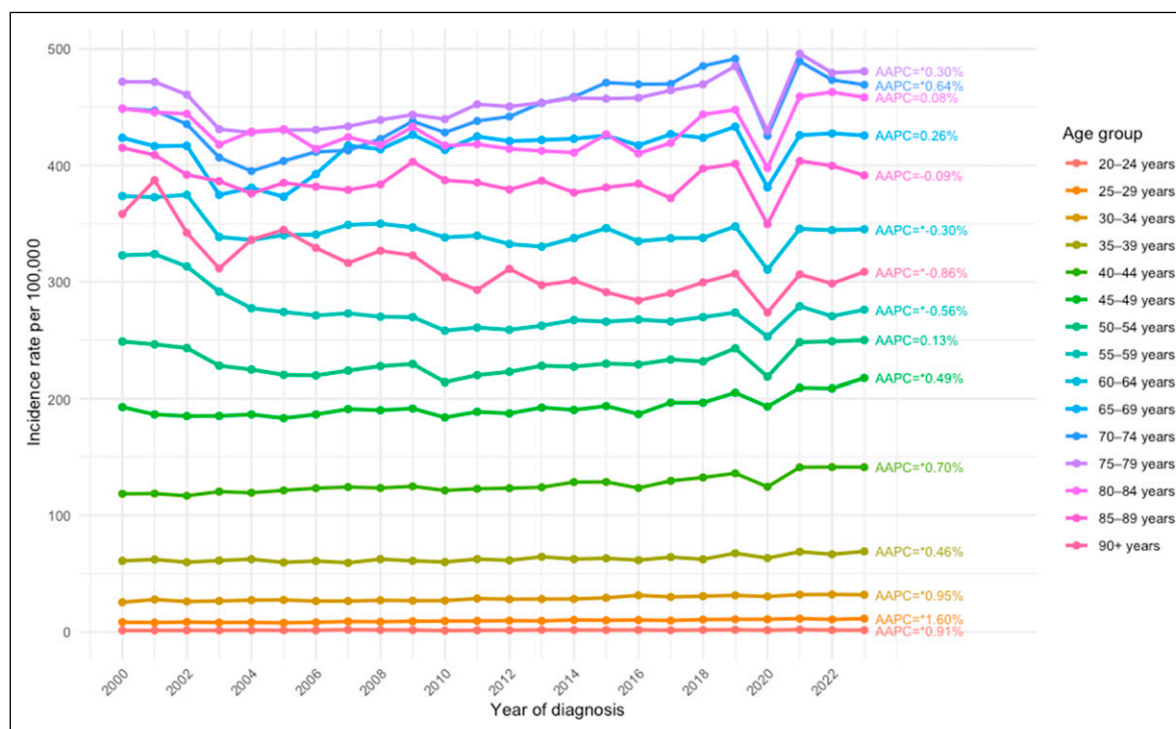


Figure 1. Joinpoint regression analysis by age-groups. Figure 1 shows breast cancer incidence rates (per 100,000) by 15 age groups from 2000 to 2023, based on joinpoint regression analysis. A clear and intuitive pattern is revealed: the older the age group, the higher the incidence rate

partly reflect better detection, increasing exposure to risk factors, such as delayed childbearing and Western dietary habits, likely play a significant role.²⁹ Additionally, aggressive tumor subtypes (such as triple-negative breast cancer), later-stage diagnosis, and unequal access to treatment continue to contribute to the highest breast cancer mortality rates observed among non-Hispanic Black women, despite comparable or even lower incidence rates.^{2,3}

Public Health and Policy Implications

Public health screening programs are directly impacted by the increased prevalence of breast cancer, especially among younger women and minority groups. The U.S. Preventive Services Task Force (USPSTF) recently updated its recommendations to suggest that average-risk women begin biennial mammograms at age 40 in recognition of the consistent rise in breast cancer among women in their 40s.³⁰ Our findings, which show the rising incidence of early-onset breast cancer among a population that was previously ineligible for routine screening, provide credence to this policy change.

However, expanding screening alone is not sufficient. A significant concern is the difference in mammography access by race, region, and socioeconomic status.²¹ To ensure equitable access to these screening services, targeted interventions and community-based preventative programs must be implemented.²¹ The anticipated increase in breast cancer cases, due to both incidence trends and demographic aging, also necessitates investment in oncology infrastructure, patient navigation services, and survivorship care planning.^{3,28}

Strengths and Limitations

The use of extensive, population-based data spanning more than 20 years is a key strength of this study, as it enables reliable trend analysis by age and race/ethnicity. This period also captures the post-Women's Health Initiative period, the post-2004 incidence rebound, and the COVID-19 pandemic disruption. We simultaneously stratified trends by age, race/ethnicity, and gender using Joinpoint regression, allowing identification of specific time points within each subgroup, which is not provided by surveillance data from professional organizations. In addition, we explicitly addressed the COVID-19 pandemic effects on incidence trends, including a sensitivity analysis excluding the years 2020 and 2021, years substantially affected by COVID-19-related screening delays, and re-running the Joinpoint regression on the 2000-2019 period to present the estimates.

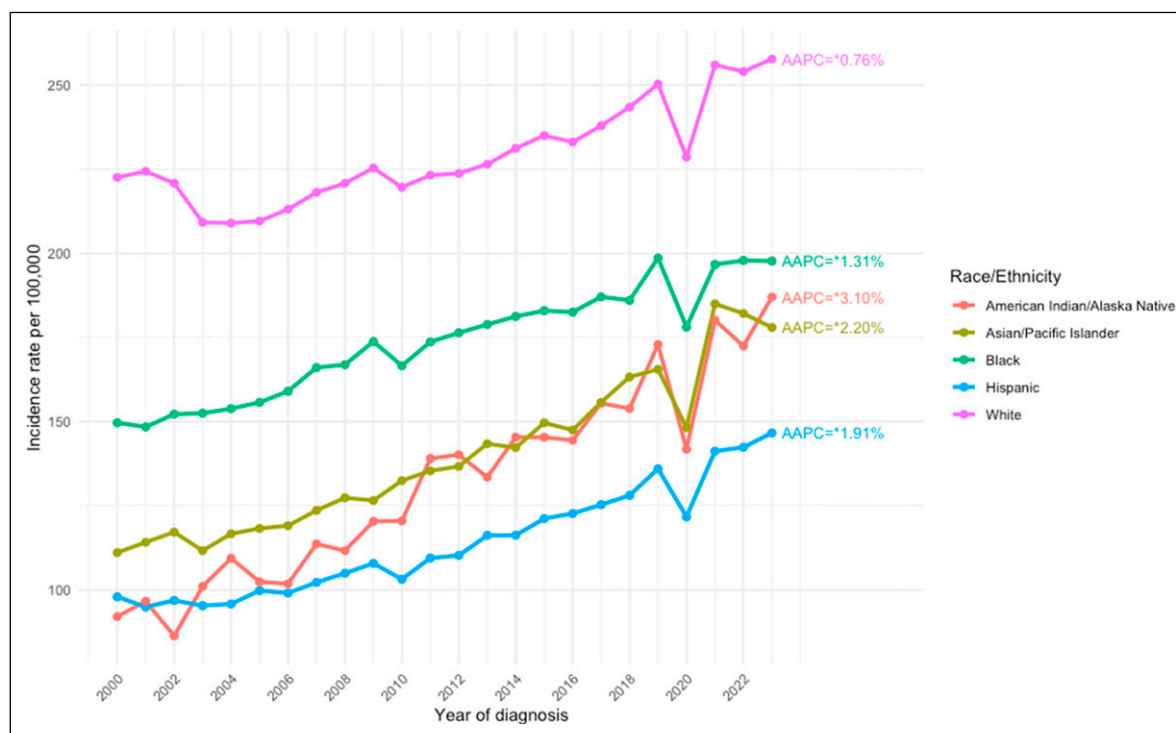


Figure 2. Joinpoint regression analysis by race/ethnicity. Figure 2 shows breast cancer incidence rates (per 100,000) by race/ethnicity from 2000 to 2023, based on joinpoint regression analysis

Nonetheless, several limitations must be noted in this study. Since it is an ecological analysis, our study cannot establish a causal relationship between risk variables at the individual level. Furthermore, our study did not stratify by stage or tumor subtype (e.g., hormone receptor status), which can affect incidence and results. Temporal shifts in screening procedures may have also impacted case detection, especially during the COVID-19 epidemic. During the early pandemic, routine breast cancer screening programs were widely suspended by healthcare facilities. For example, in late March 2020, a joint statement by the American Society of Breast Surgeons (ASBrS) and American College of Radiology (ACR) recommended postponing all breast screening exams. As a result, far fewer breast cancers were detected in 2020 than expected under normal circumstances. This temporary deficit in case detection means that some cancers that would have been diagnosed in 2020 were likely delayed until 2021 or later, potentially at more advanced stages.

These pandemic-driven anomalies in 2020 expose challenges for long-term trend analysis. Including the 2020 incidence data point in calculations can distort temporal trends and bias AAPC estimates. Joinpoint regression models, which assume smooth trend segments, are not designed to accommodate such a one-time abrupt drop in incidence.¹² Any estimates of AAPC that include the pandemic years may be biased or unreliable, and we acknowledge that our reported trend for the late study period may partly capture this artifact rather than a sustained change in incidence.

Despite these limitations, our analysis provides valuable insights into breast cancer incidence dynamics over two decades. The results, especially those during the COVID-19 era, should be interpreted in light of the extraordinary healthcare disruptions. We recommend continued monitoring in the coming years to determine if the post-pandemic data realign with pre-pandemic trends or if lasting effects such as a rebound or shift in incidence emerge once routine screening and medical care are normalized.

Conclusion

In conclusion, the current study of the incidence of breast cancer in the US (2000–2023) reveals a gradual but noticeable increase in rates over time, which translates into a significant and constantly expanding cancer burden. While overall rates have increased, age- and race-specific patterns highlight important heterogeneity in these trends. The incidence of breast cancer is rising significantly among younger women, while it first declined and then stabilized among older women. Simultaneously, racial and ethnic differences in incidence are evolving, with historically lower-incidence groups showing significant increases and narrowing the gap with White women. These findings have important implications for cancer

control. They underscore the need for adaptive screening strategies, including recent policy changes recommending mammography initiation at age 40, and sustained investment in preventive health measures and healthcare resources to manage the expanding burden.

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Ethical Considerations

The study used de-identified, publicly available data from the National Cancer Institute's Surveillance, Epidemiology, and End Results (SEER) program. Formal ethical approval was not required as no individual patient data or identifiable information was accessed. Informed consent was not applicable for the study.

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Data Availability Statement

All data analyzed during this study are included in this published article.

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